

THE WITNESS

ANALYSIS

A 2D top-down stealth game



Client	Nightfall Interactive
Prepared by	David Eslava
Programme	Fontys ICT — StartSemester, Challenge 1
Document	Analysis — phase 1 of 5
Version	0.2 - 22 September 2026

Contents

1. Introduction.....	3
1.1 Purpose of this document.....	3
1.2 The client	3
1.3 The product in one paragraph	3
2. Problem definition	3
2.1 The client's problem.....	3
2.2 The project challenge	3
2.3 Target audience.....	4
3. Scope	4
3.1 In scope.....	4
3.2 Out of scope	4
3.3 Designed, but not built in this delivery	4
4. Requirements (MoSCoW)	5
M Must Have	5
S Should Have	5
C Could Have	6
W Will Not Have	6
5. Proposed solution	7
6. Risks	7
7. Sources	8
8. Document history.....	8

1. Introduction

1.1 Purpose of this document

This document describes the analysis phase of The Witness, a 2D top-down stealth game commissioned by Nightfall Interactive. It defines the problem the client wants to solve, the scope of the project, and the requirements the delivered product must meet. It does not describe how the product will be built; the technical approach is documented separately in the Design document, and the schedule in the Project Plan.

1.2 The client

Nightfall Interactive is a small independent game studio that publishes short, stylised games. The studio wants to test whether a learning enemy that visibly gets better at hunting the player is a strong enough hook to build a real game around. Before committing to a full production budget, the studio asked me for a playable prototype that demonstrates the concept.

1.3 The product in one paragraph

The Witness is a 2D top-down stealth game set in a 1960s jazz club. The player character witnessed something they were not supposed to see, a murder or a dirty deal, and must escape the building before the gangsters inside catch them. The game combines two dynamics: **stealth**, where ordinary guards patrol fixed routes and detect the player through their line of sight, and **pursuit**, where the special enemy The Hunter, the boss's right-hand man, is designed to learn over time to intercept the player more efficiently instead of following a fixed pattern.

2. Problem definition

2.1 The client's problem

Stealth games traditionally rely on enemies driven by fixed, hand-written rules. Once the player understands the rules, the tension disappears, the enemy becomes a puzzle with a known solution, and replay value drops significantly. Nightfall Interactive wants to know whether an enemy that adapts to the player can keep the tension alive across repeated sessions, and whether that adaptation is visible enough to the player to be worth marketing.

2.2 The project challenge

The studio agreed to do this in two steps. A learning enemy can only be judged if there is a working stealth game around it, so the first delivery has to establish that foundation and produce a worked-out design for the enemy itself.

Deliver a playable prototype that demonstrates the stealth loop in a single short level, and a complete design for the adaptive enemy, including the method that would be used to measure whether it improves.

The challenge has two halves that must both succeed:

- **The game must be playable.** The stealth loop, move, hide, be seen, get caught, must work on its own, otherwise there is nothing for the adaptive enemy to be measured against.

- **The design must be specific enough to build from.** A description of an enemy that learns is worthless if it does not say what the enemy perceives, what it can do, and how its improvement would be measured.

2.3 Target audience

Aspect	Description
Primary players	Players aged 16+ who enjoy short, tense, single-session stealth games
Familiarity assumed	Comfortable with keyboard movement controls. No prior experience with stealth games required
Session length	3 to 6 minutes per attempt, designed for repeated attempts
Platform	Desktop (Windows / Linux), keyboard input

3. Scope

3.1 In scope

- One small playable level set inside the jazz club
- A player character that moves and cannot walk through walls
- Guards that patrol and see the player through a vision cone
- A game over screen when the player is caught
- A complete design for The Hunter, the learning enemy, including how its improvement would be measured

3.2 Out of scope

- More than one level, or a story with an ending
- Finished art, voice acting or licensed music
- Console or mobile versions
- Online or multiplayer
- Saving your progress

3.3 Designed, but not built in this delivery

The Hunter's learning behaviour is fully designed in this project but is not implemented in the prototype. The available time went into the stealth loop, which has to work before any learning behaviour can be measured against it. What The Hunter perceives, the moves it can make, how it is rewarded, and the method for measuring whether it improves are all documented in the Design document. Building and measuring it is the first recommendation for the next iteration.

4. Requirements (MoSCoW)

The requirements are prioritised using MoSCoW, a method that sorts them into four groups, from "the project fails without this" down to "we are deliberately not doing this". Each requirement is written as a user story, which names who wants it, what they can do, and why it matters, so the table stays readable for someone who does not know the project.

M Must Have

Without these there is nothing to demonstrate.

ID	User story
M-01	As the client, I can start the game and play it myself, so that I can judge the concept instead of reading about it
M-02	As a player, I can move my character in eight directions with the W, A, S and D keys, so that I can navigate the club
M-03	As a player, I cannot walk through walls or furniture, so that the level behaves like a real space
M-04	As a player, I can use walls and furniture as cover, so that where I stand actually matters
M-05	As a player, I meet enemies that move through the level, so that there is something to avoid
M-06	As a player, I can only be seen when I am inside an enemy's vision cone, so that hiding behind cover actually works
M-07	As a player, I see enemies walk around walls instead of into them, so that their movement is believable
M-08	As a player, I am told the attempt is over when an enemy catches me, so that being spotted has a consequence

S Should Have

Strongly wanted and they add real value, but the prototype still works without them.

ID	User story
S-01	As a player, I can sprint for a few seconds using a stamina bar that refills when I stop, so that I can escape a bad moment without being able to run forever
S-02	As a player, I can watch a meter fill while an enemy looks at me, so that I have a moment to break the line of sight before I am caught
S-03	As a player, I can see an enemy search the place it last saw me before returning to its patrol, so that escaping feels earned
S-04	As a player, I can start from a main menu and retry from a game over screen, so that I can make repeated attempts without restarting the program

C Could Have

Included only if time remains once the Must and Should requirements are met.

ID	User story
C-01	As a player, I can face an enemy that gets better at chasing me the more I play, so that the game stays tense after I have learned the guards' routes
C-02	As a player, I can hide under a table or behind the bar while I stay still, so that I have somewhere to go when a guard approaches
C-03	As a player, I can throw or activate an object to pull a guard off their route, so that I can create an opening
C-04	As a player, I can hear the audio react while a guard is searching, so that I feel the danger without watching the interface
C-05	As a player, I can fight an enemy instead of avoiding them, so that being cornered is not always fatal

C-01 is the client's core research question. It is fully designed in this project, including how its improvement would be measured, but it is not implemented in this prototype. The stealth loop must work before any learning behaviour can be measured against it.

W Will Not Have

Explicitly excluded from this project, so expectations are clear.

ID	Requirement	Why it is excluded
W-01	Multiple levels or vertical floors	One level is enough to prove the concept. More levels multiply the work without adding evidence
W-02	Multiplayer	Out of scope for a single-player stealth concept and far beyond the available time
W-03	Advanced settings menu	Audio and graphics options add no value to a prototype that is demonstrated, not shipped
W-04	Crafting systems	Belongs to a different genre and does not support the research question
W-05	Adaptive jazz soundtrack	Attractive for a full title, but licensing and implementation cost cannot be justified here

5. Proposed solution

One floor of the jazz club: a main room with tables and a bar, a back corridor, and an exit the player is trying to reach. Small enough that one attempt takes a few minutes.

The guards are predictable on purpose. They are the comparison. The Hunter is the interesting one: instead of following a route, he decides where to move based on where the player is relative to him, and he adjusts those decisions based on what happened in earlier attempts. Moves that brought him closer get reinforced, moves that did not are discouraged. Over many attempts his route to the player should get more direct. This behaviour is designed in this project but not implemented in the prototype.

Because the client's question is whether this actually works, the design specifies what the prototype would need to record per attempt, for example how long The Hunter needed to reach the player. That is what would make it possible to compare a trained Hunter against an untrained one. The measurement method is described in the Design document.

The game uses a high contrast noir look with a very limited colour palette, so the player can instantly read the space, the enemies and the vision cones. The reasoning behind that choice and the tools I picked are in the Design document.

6. Risks

- **The stealth loop could take longer than expected.** It is the part everything else depends on, so it is built first with placeholder graphics, before any visual work.
- **The scope could be too big for four weeks on my own.** I deal with this by finishing all the Must Have requirements before starting anything else, in the order listed in section 4.
- **This is my first project in a game engine, so problems will take longer than I estimate.** Using placeholder graphics until the gameplay works keeps learning the engine from blocking the core game.
- **The Hunter could turn out to be harder to build than its design suggests.** Because it is not implemented in this delivery, that risk is carried into the next iteration rather than resolved here. The design keeps what The Hunter perceives and the moves it can make deliberately small, to keep that risk low.

7. Sources

These are the sources I used during the analysis. They fall into three groups: the reinforcement learning research for *The Hunter*, the research into which engine to use and how to work with it, and the templates I looked at to learn how a game design document is normally written. All of them were consulted between 1 and 22 September 2026.

Reinforcement learning and AI — to decide which algorithm would suit *The Hunter*:

- [Deep Reinforcement Learning Applied to the Snake Video Game](#)
- [Q Learning Intro/Table - Reinforcement Learning p.1](#)
- [Search basic game applying reinforcement learning](#)
- [Deep Reinforcement Learning Tutorial for Python in 20 Minutes](#)
- [The FASTEST introduction to Reinforcement Learning on the internet](#)
- [A.I. Learns to Drive](#)
- [A Beginner's Guide to Q-Learning: Understanding with a Simple Gridworld Example](#)

Engine research — choosing an engine and learning to work with Godot:

- [How to Make a Game in Python](#)
- [Making a Game in Python with No Experience](#)
- [Godot 4 Crash Course #1 - The Basics](#)
- [How to make a Video Game - Godot Beginner Tutorial](#)
- [True Top-Down 2D](#)

Game design document templates — to learn how this kind of document is normally structured

- [Detailed Game Design Documentation Template](#)
- [How to write an effective Game Design Document \(GDD\)?pls help](#)
- [How to Write a Game Design Document](#)
- [A GDD Template for the Indie Developer](#)
- [A Template for Analyzing Game Design.](#)

8. Document history

Version	Date	Change
0.1	06-09-2026	First version: concept, MoSCoW table and proposed solution
0.2	22-09-2026	Rewritten for the client after the coach review of 14-09-2026. Technical choices moved to the Design document, Requirements rewritten as user stories. Scope updated: <i>The Hunter</i> is designed but not built in this delivery